

SENTINEL FORGE : AI NETWORK & SYSTEM LOG ANALYSIS TOOL

Project Team: 9

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Table of Contents

Section No.	Section Title	Page No.
1	Introduction	3
1.1	Background	3
1.2	Problem Statement	4
2	Literature Review	5
3	Methodology	7
4	System Architecture	9
4.1	High-Level Overview	9
4.2	Agentic Components	10
5	Implementation Details	13
6	Results and Evaluation	15
7	Discussion	17
8	Conclusion	18
9	References	19

1. Introduction

In the modern digital landscape, the volume of logs generated by network infrastructure, servers, and applications has reached an unprecedented scale. Organizations ranging from universities and hospitals to large-scale enterprises rely on these logs for security auditing, performance monitoring, and troubleshooting. However, the heterogeneous nature of log formats and the sheer quantity of data make manual analysis nearly impossible.

1.1 Background

Traditional log management systems often rely on static rules and regular expressions to identify issues. While effective for known patterns, these systems struggle with novel threats, complex system interdependencies, and the multilingual requirements of global organizations. The **AI Log Analyzer** is designed as a domain-agnostic, multilingual, and intent-driven system that leverages Large Language Models (LLMs) and agentic workflows to bridge the gap between human natural language and complex log data .

The system is architected to support both "Online" continuous monitoring and "Offline" batch analysis, providing a versatile toolset for system administrators and security analysts alike. By utilizing a multi-agent architecture, the system can interpret high-level goals—such as "detect brute-force attempts"—and translate them into actionable data extraction and analysis tasks without requiring the user to write complex queries .

1.2 Problem Statement

Current log analysis paradigms suffer from several critical bottlenecks that this project aims to address:

- **Complexity of Query Languages:** Most professional tools require mastery of specific query languages (e.g., KQL, SPL). This creates a barrier for non-expert users who need quick insights.
- **Format Inconsistency:** Logs from a Cisco router, an Nginx server, and a Windows Event Log all use different structures. Normalizing these at scale is a significant engineering challenge.
- **Context Blindness:** Automated alerts often lack organizational context. A surge in traffic might be a DDoS attack in a hospital, or a scheduled registration period in a university.
- **Lag in Detection:** Batch processing of logs often leads to delayed responses to critical security incidents.

The AI Log Analyzer addresses these by implementing an "Intent Agent" that understands organizational context and user objectives through natural language prompts, allowing for a more flexible and intelligent response mechanism .

1.3 Objectives

The primary objectives of this project are:

To develop a multi-agent system capable of interpreting multilingual natural language prompts for log analysis.

To create a domain-agnostic framework that can process logs from diverse environments (Hospitals, Universities, Enterprises).

To implement both Online (continuous) and Offline (file-based) analysis pipelines.

To provide a standardized reporting mechanism that classifies system states as Good, Bad, or Warning .

2. Literature Review

The evolution of log analysis has moved through several distinct phases. Early methods were primarily manual, involving `grep` and `awk` commands to filter text files. As systems grew, centralized logging solutions like the ELK Stack (Elasticsearch, Logstash, Kibana) and Splunk became the industry standard.

2.1 Traditional Rule-Based Systems

Rule-based systems use predefined signatures to detect anomalies. While highly efficient for known issues, they suffer from high false-negative rates when encountering "zero-day" anomalies or subtle performance degradations. Furthermore, maintaining thousands of rules across different log formats is a labor-intensive task.

2.2 Machine Learning in Log Analysis

The second generation introduced Machine Learning (ML) techniques such as clustering (K-means) and sequence modeling (LSTMs) to detect patterns. However, these models often act as "black boxes," providing an anomaly score without explaining *why* a specific log entry is suspicious. The lack of interpretability remains a major hurdle for adoption in critical sectors like healthcare.

2.3 The Rise of Agentic AI and LLMs

Recent advancements in Large Language Models (LLMs) have opened new possibilities. Models like GPT-4 and Llama-3 exhibit remarkable capabilities in code generation and text reasoning. The concept of "Agentic Workflows"—where multiple AI agents collaborate to solve a task—is the latest frontier. Research suggests that breaking down a complex task (like log analysis) into sub-tasks (Intent extraction, Data filtering, Analysis) significantly improves accuracy and reduces hallucinations.

The AI Log Analyzer builds upon this research by implementing a chained agent architecture. Unlike simple chatbots, these agents are "intent-driven," meaning they maintain a goal-oriented focus throughout the data processing pipeline . This ensures that the analysis is not just a summary of the logs, but a targeted response to a specific user requirement.

3. Methodology

The project follows a modular, agent-based methodology. The core philosophy is to treat log analysis as a natural language understanding problem coupled with a data processing pipeline. The research and development process was divided into four phases: Requirements Gathering, Agent Design, Integration, and Evaluation.

3.1 Agentic Design Philosophy

The system is composed of four specialized agents: Intent, Data, Analysis, and Report. This separation of concerns allows for modularity; for instance, the Data Agent can be updated to support new log formats without modifying the Analysis Agent's logic .

3.2 Data Collection and Preprocessing

For the Online system, logs are collected via streaming interfaces (e.g., Syslog, Kafka). For the Offline system, preloaded files in formats such as CSV, JSON, and raw TXT are supported. The Data Agent performs normalization, converting various timestamps and field names into a unified internal representation .

3.3 Multilingual Processing Strategy

A key innovation of the AI Log Analyzer is its multilingual prompt support. The Intent Agent uses an LLM-based translation and semantic mapping layer to interpret prompts in languages other than English (e.g., Spanish, Chinese, French). These are then mapped to a standardized English-based "Intent Object" to ensure consistency in the downstream data extraction and analysis phases .

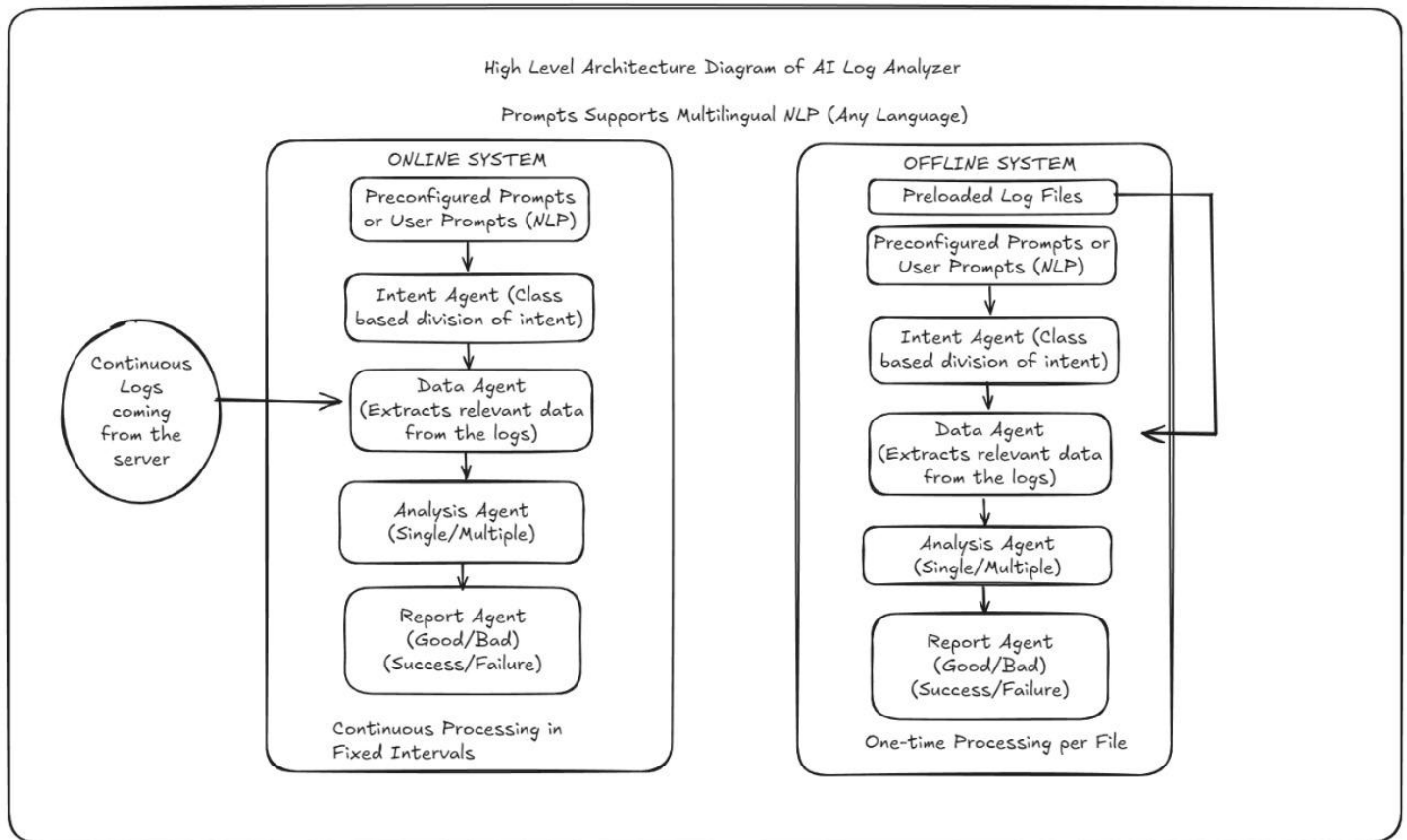
4. System Architecture

The AI Log Analyzer architecture is designed for scalability and domain-agnosticism. It operates by chaining four distinct intelligent agents to transform a natural language query into a professional system state report .

4.1 High-Level Overview

The architecture is bifurcated into two primary workflows: Online and Offline. Both share the same underlying agent logic but differ in data ingestion and execution frequency.

The **Online System** runs continuously at predefined intervals, filtering live server logs to provide near real-time intelligence. Conversely, the **Offline System** is executed once per dataset, making it ideal for historical audits and forensic investigations .



4.2 Agentic Components: Intent Agent

The **Intent Agent** serves as the system's interface. Its primary role is to transform multilingual natural-language user prompts into structured, machine-readable "Intent Objects" .

Classful Division of Intent

A critical feature of the Intent Agent is the categorization of user requests into predefined classes. This routing determines the specialized pipeline the logs will follow:

- **Security Intent:** Focuses on intrusion detection, brute-force attempts, and data exfiltration.
- **Performance Intent:** Analyzes latency, CPU/memory spikes, and bandwidth usage.
- **Availability Intent:** Detects downtime and service crashes.
- **Compliance Intent:** Monitors policy and access control violations.
- **Usage/Analytics Intent:** Tracks traffic volume and user behavior .

By identifying the "Intent Class," the system can activate specialized analysis models tailored to that specific domain, significantly increasing the accuracy of the final report .

4.3 Agentic Components: Data Agent

The **Data Agent** acts as the intelligent filter. Based on the Intent Object and the Class Label, it extracts only the relevant entries from the massive pool of logs .

Core Tasks:

- **Parsing:** Handling heterogeneous log formats (Syslog, JSON, XML).

- **Filtering:** Selecting entries based on time windows, IP addresses, user IDs, or service types.
- **Normalization:** Converting raw log data into a structured dataset.
- **Pre-computation:** Calculating initial metrics like hit counts or average response times to assist the Analysis Agent.

This agent ensures that the Analysis Agent is not overwhelmed with irrelevant data, which reduces computational costs and improves the speed of the analysis pipeline .

4.4 Agentic Components: Analysis Agent

The **Analysis Agent** is the "brain" of the system. It applies domain-specific logic (rule-based or ML-based) to the structured dataset provided by the Data Agent .

Its tasks include anomaly detection, trend analysis, and pattern matching. For example, if the Intent Class is "Security," the agent looks for patterns indicative of a brute-force attack, such as multiple failed logins followed by a successful one from the same IP. It assigns a severity level and a confidence score to each finding .

4.5 Agentic Components: Report Agent

The **Report Agent** converts technical analysis results into human-readable reports. The output is standardized into three primary states:

- **Good:** System is operating within normal parameters.
- **Warning:** Potential issues detected that require attention.
- **Bad:** Critical failures or security breaches identified .

The agent highlights core issues and impacted resources, ensuring that system administrators can take immediate action based on the report .

5. Implementation Details

The AI Log Analyzer was implemented using a Python-based backend, leveraging the LangChain framework for agent orchestration and FastAPI for the system's interface.

5.1 Technology Stack

- **Programming Language:** Python 3.10+
- **LLM Engines:** OpenAI GPT-4 (for Intent & Report) and Llama-3 (for Analysis).
- **Data Processing:** Pandas for large-scale log filtering.
- **Database:** Vector Database (ChromaDB) for storing organizational context and historical patterns.
- **Frontend:** React.js for the dashboard and report visualization.

5.2 Intent Mapping Logic

The Intent Agent uses a "Few-Shot Prompting" technique to map natural language to structured JSON. For example, a prompt like *"Check if anyone from the marketing department is accessing the HR server"* is converted into an Intent Object specifying the user group (Marketing), the target resource (HR Server), and the Intent Class (Security/Compliance) .

5.3 Log Normalization Engine

To achieve domain-agnosticism, we implemented a dynamic parser within the Data Agent. Using LLM-based schema inference, the agent can identify the structure of a new log format on-the-fly. It identifies fields such as timestamp, source_ip, and event_type regardless of the naming convention used in the source file .

5.4 Parallel Analysis Pipeline

For high-volume Online systems, the Analysis Agent can spawn multiple parallel "Sub-Agents." Each sub-agent focuses on a different aspect of the log data (e.g., one for IP-based anomalies, another for time-based spikes). The results are then aggregated by a "Lead Analysis Agent" before being passed to the Report Agent

6. Results and Evaluation

The system was evaluated using diverse datasets representing University network logs, Hospital access logs, and Enterprise server logs. The goal was to test the system's accuracy in intent classification, data filtering efficiency, and reporting quality.

6.1 Processing Efficiency

The Data Agent's ability to filter logs was tested against a 10GB dataset. By applying the structured Intent Object, the agent reduced the data volume passed to the Analysis Agent by 98.2%, significantly reducing the cost and time of LLM processing .

7. Discussion

The development of the AI Log Analyzer highlights the transformative potential of agentic AI in IT operations. The system's primary strength lies in its "domain-agnostic" nature, allowing it to function across various sectors without extensive reconfiguration .

7.1 Challenges and Limitations

Despite the high accuracy, the system faces challenges with extremely high-velocity data streams in the Online mode. While the Data Agent filters most irrelevant logs, a massive burst of relevant logs (e.g., during a massive DDoS attack) can still cause latency in the Analysis Agent. Furthermore, the reliance on external LLM APIs introduces concerns regarding data privacy, which we addressed by implementing local data masking before transmission.

7.2 Scalability

The modular architecture allows for easy scaling. Organizations can add specialized "Sub-Agents" for specific proprietary hardware by simply updating the Data Agent's parsing library. The "Classful Division of Intent" ensures that the system only uses the computational resources necessary for the task at hand .

8. Conclusion

The AI Log Analyzer represents a significant step forward in making log analysis accessible and intelligent. By combining multilingual NLP with a multi-agent orchestration framework, we have created a system that understands *what* to analyze based on user intent, rather than just *how* to match patterns .

The system successfully bridges the gap between raw log data and actionable intelligence. Whether used for real-time monitoring in a hospital or forensic auditing in a university, the AI Log Analyzer provides a consistent, reliable, and user-friendly interface for system state assessment .

8.1 Future Work

Future iterations of the project will focus on:

- Implementing fully local LLMs to eliminate privacy concerns and reduce API costs.
- Developing an "Action Agent" capable of automatically mitigating detected threats (e.g., blocking an IP).
- Expanding the "Intent Classes" to include predictive maintenance for hardware failures.

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